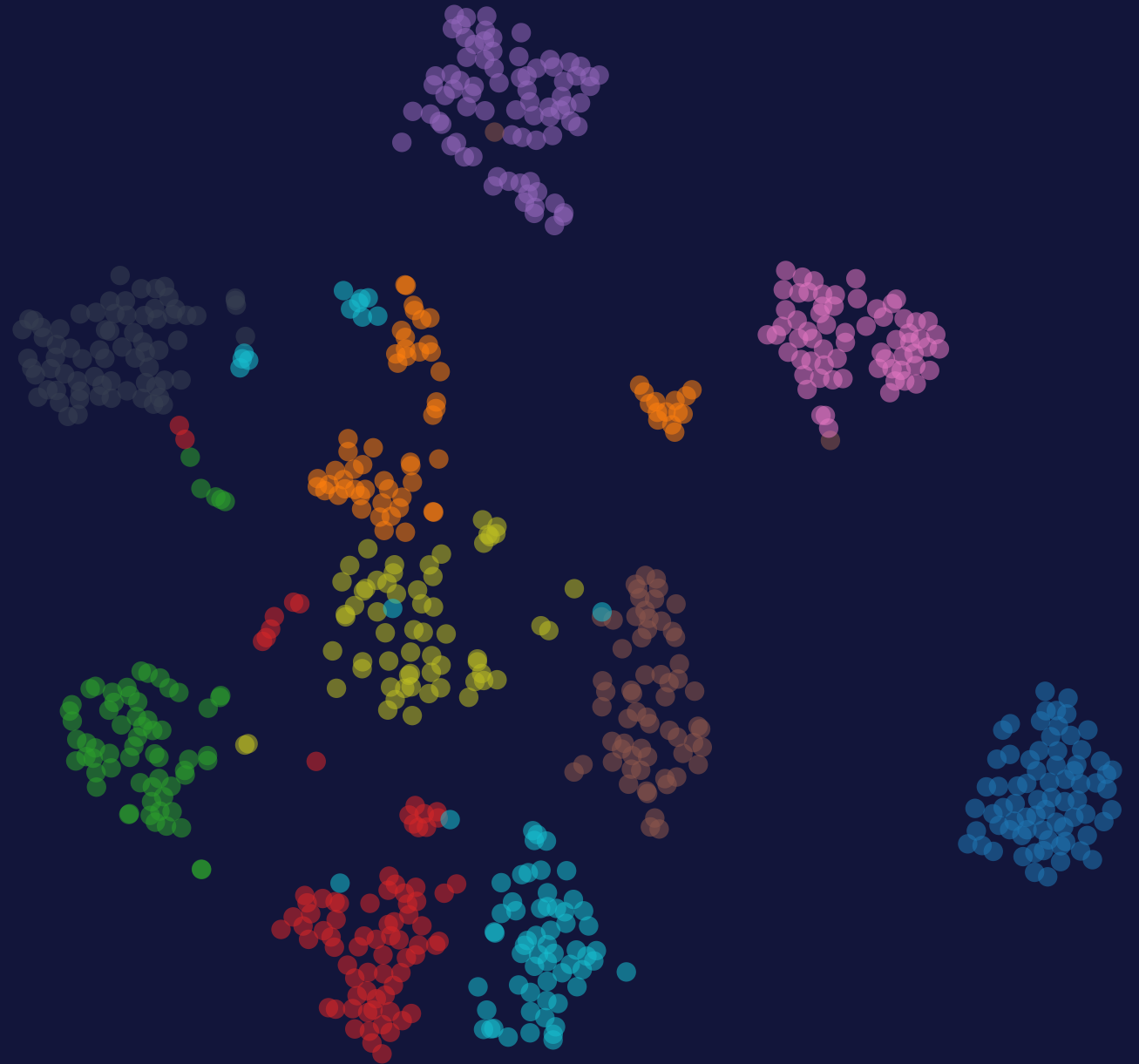


t-SNE

t-distributed Stochastic Neighbor Embedding

How a computer turns hundreds of numbers into one picture you can see.



What we will cover

1

Why do we need it?

- Data with too many columns
- What PCA does well
- Where PCA struggles
- Spectral clustering idea

2

How does it work?

- 7 simple steps
- Friends, probabilities, curves
- Push and pull
- A live run, round by round

3

How do we read it?

- What we can trust
- What we should not trust
- The perplexity setting
- Practical tips

THE PROBLEM

One picture is many numbers

0	0	10	15	14	4	0	0
0	0	4	6	13	16	2	0
0	0	0	3	16	9	0	0
0	0	0	1	16	6	0	0
0	0	0	0	10	12	0	0
0	0	0	0	1	16	4	0
0	1	9	5	6	16	7	0
0	0	14	12	15	11	2	0

This is a handwritten 3. It is a tiny image with $8 \times 8 = 64$ pixels.

Each pixel is one number: 0 is white and 16 is dark ink.

So one picture is a list of 64 numbers.

1 picture = 64 numbers

Each number is one dimension, so this picture is a point in 64 dimensions.

THE PROBLEM

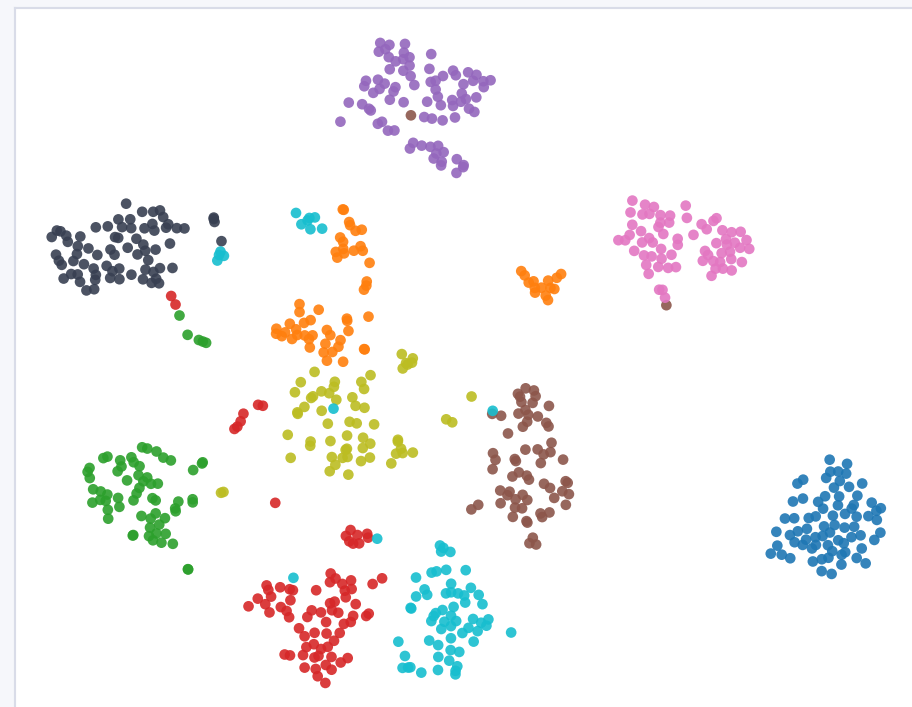
We can only draw 2 or 3 dimensions

We want a 2D map where similar pictures land close together.

Picture	p1	p2	p3	...	p64
#1	0	0	12	...	0
#2	0	0	0	...	0
#3	0	1	13	...	0
#4	0	0	2	...	0
#5	0	1	3	...	0

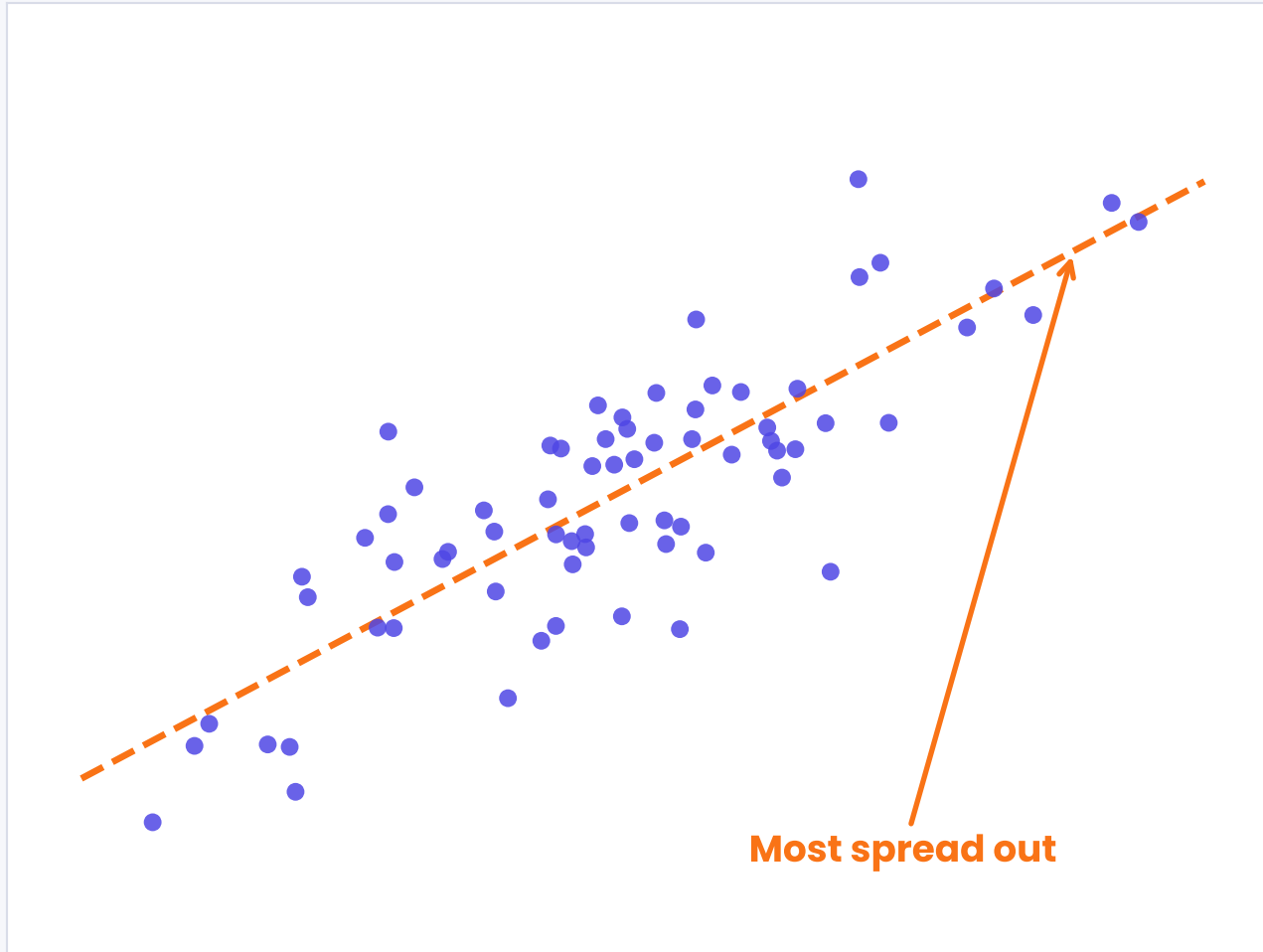
700 pictures x 64 numbers

t-SNE



2 numbers per picture = a map

PCA: the shadow method

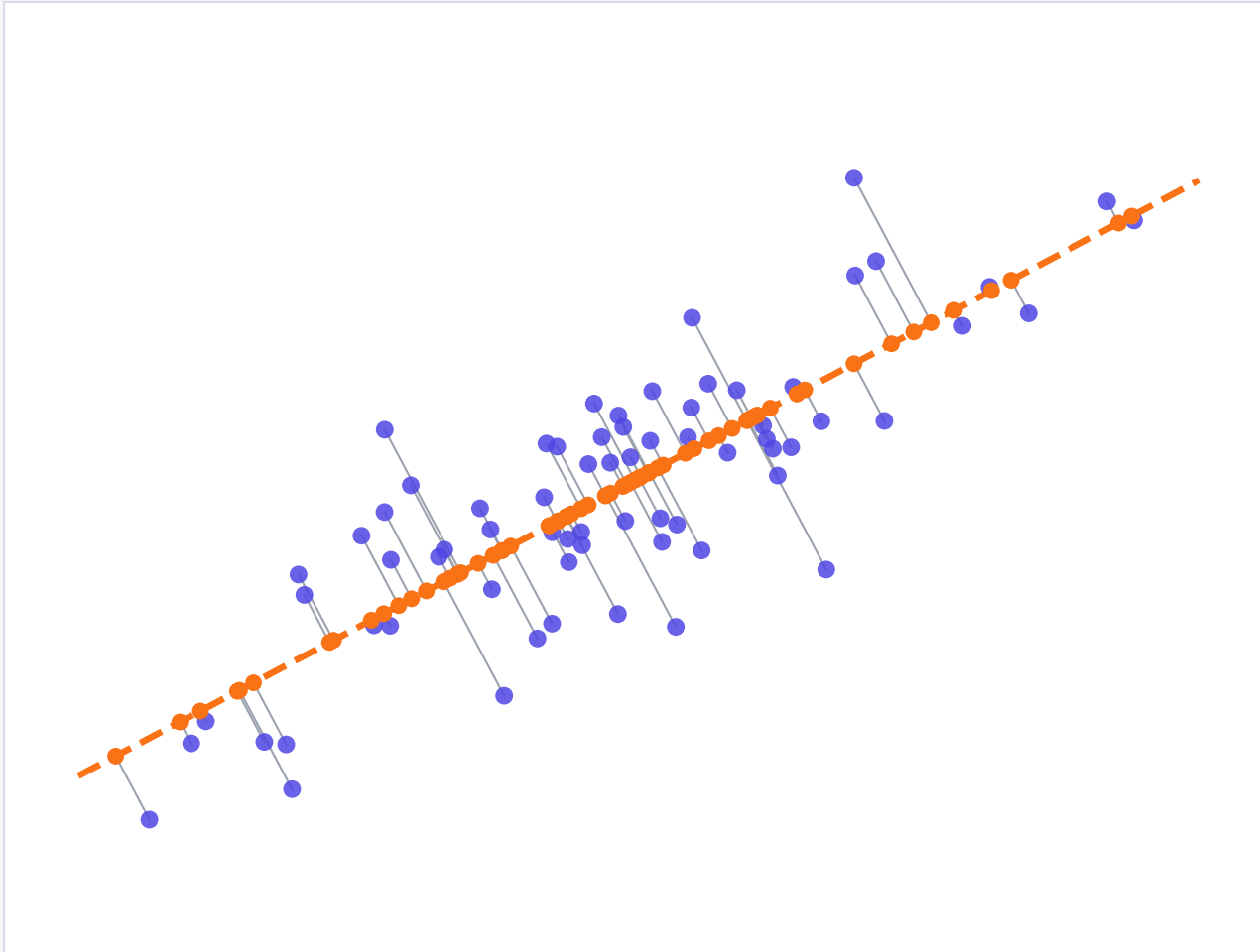


PCA means Principal Component Analysis.

It finds the straight line along which the data is most spread out.

This line keeps the most information.

PCA: the shadow method



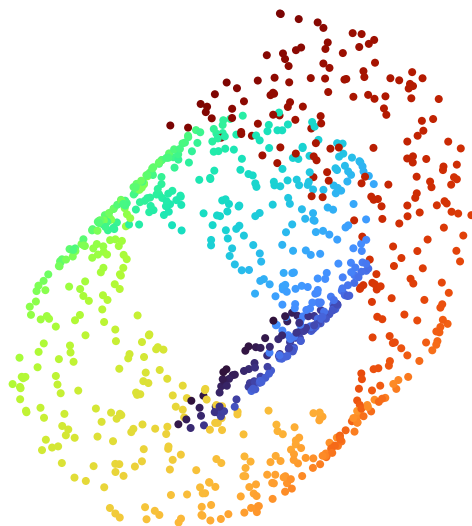
Now drop every point onto the line, like a shadow.

Each point needs only one number: its place on the line.

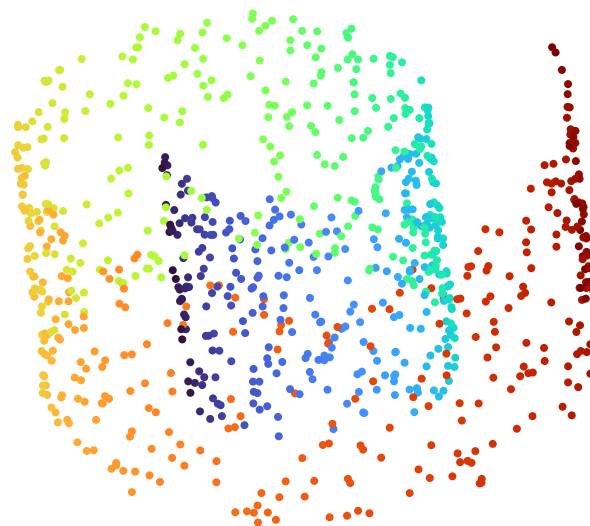
PCA is fast and simple. It works well when data follows a straight line or a flat sheet.

Where PCA struggles: curved data

1. Data: a rolled-up sheet



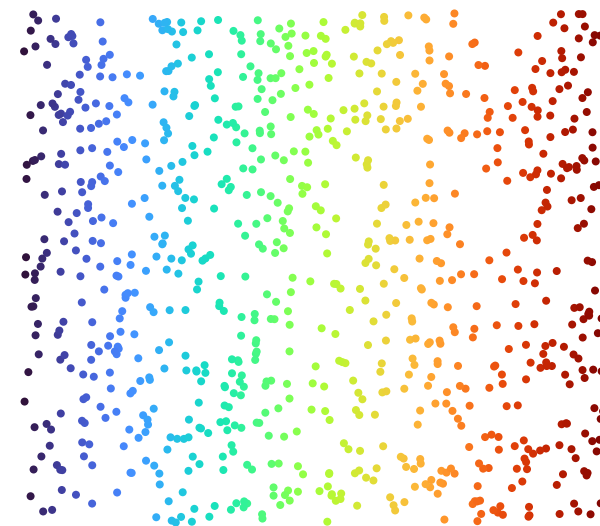
2. PCA shadow



Colors show where each point sits along the sheet.

Colors are mixed. Points from different layers land on top of each other.

3. What we wish to see



Unrolled, the colors line up. Neighbors stay neighbors.

PCA only uses straight-line shadows, so it cannot unroll a curved shape.

Same data, two maps

PCA



Groups overlap and blur together.

t-SNE



Ten clear groups, one per digit.

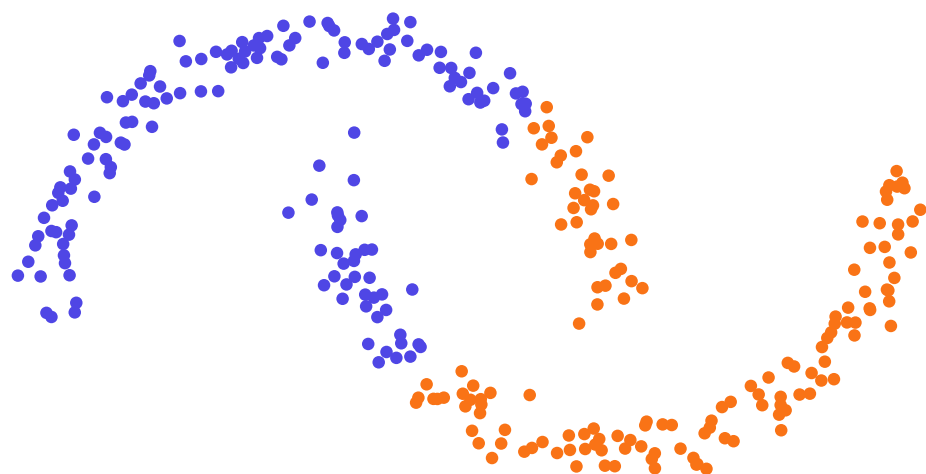
Digit:



Colors are for checking only.
Neither method used them.

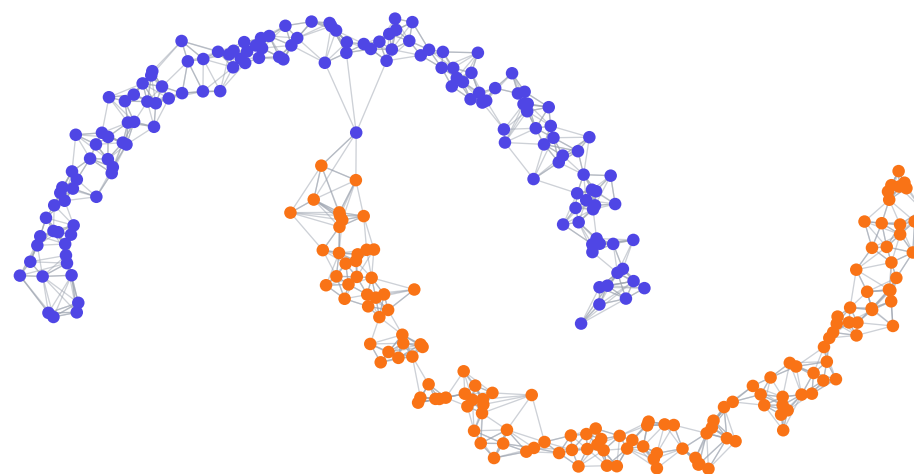
Where does spectral clustering fit in?

Straight-line thinking



A straight cut splits the moons the wrong way.

Neighbor thinking



Linking neighbors finds the two moons.

PCA thinks in straight lines. Spectral clustering links each point to its nearest neighbors. t-SNE also works with neighbors, so it can replace PCA when we want to see groups.

The big idea: keep neighbors as neighbors

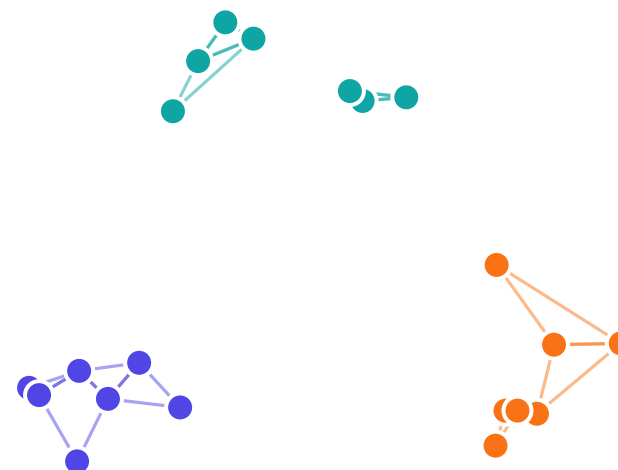
Before: friends are scattered



Friends are far apart.



After: friends sit together



Friends sit close. Strangers sit far.

Think of a seating chart: classmates who are alike sit at the same table.

The 7 steps of t-SNE

1

Measure distances

How far is each point from the others?

2

Turn them into probabilities

Close points get a high score.

3

Choose how many neighbors matter

This setting is called perplexity.

4

Build the friendship table

Combine all the scores in one table.

5

Start with a random 2D map

Scatter the points on a flat sheet.

6

Measure closeness on the map

Use a special t-curve.

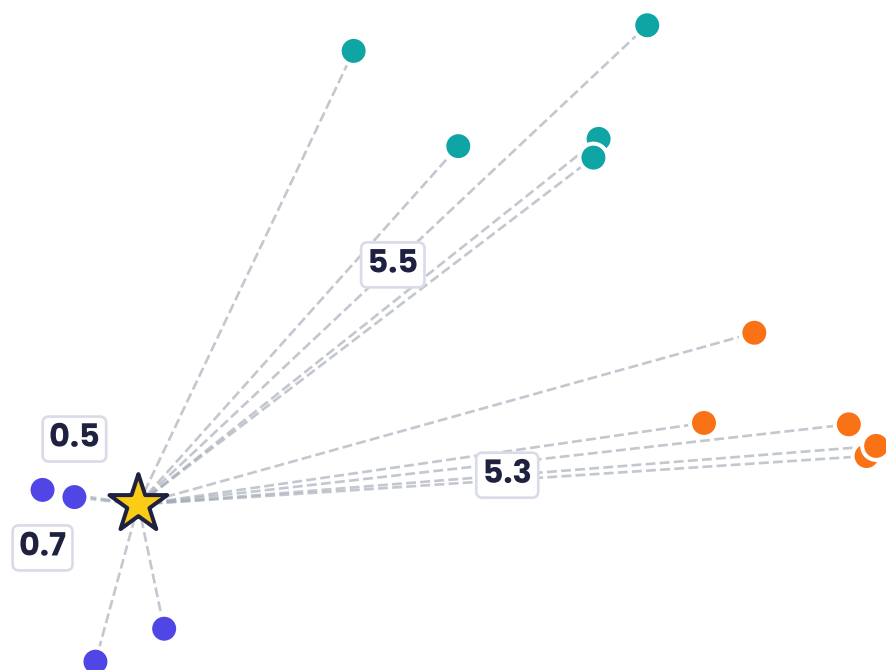
7

Push and pull, then repeat

Make the map match the table.

Then we watch it run, round by round.

Step 1: Measure distances



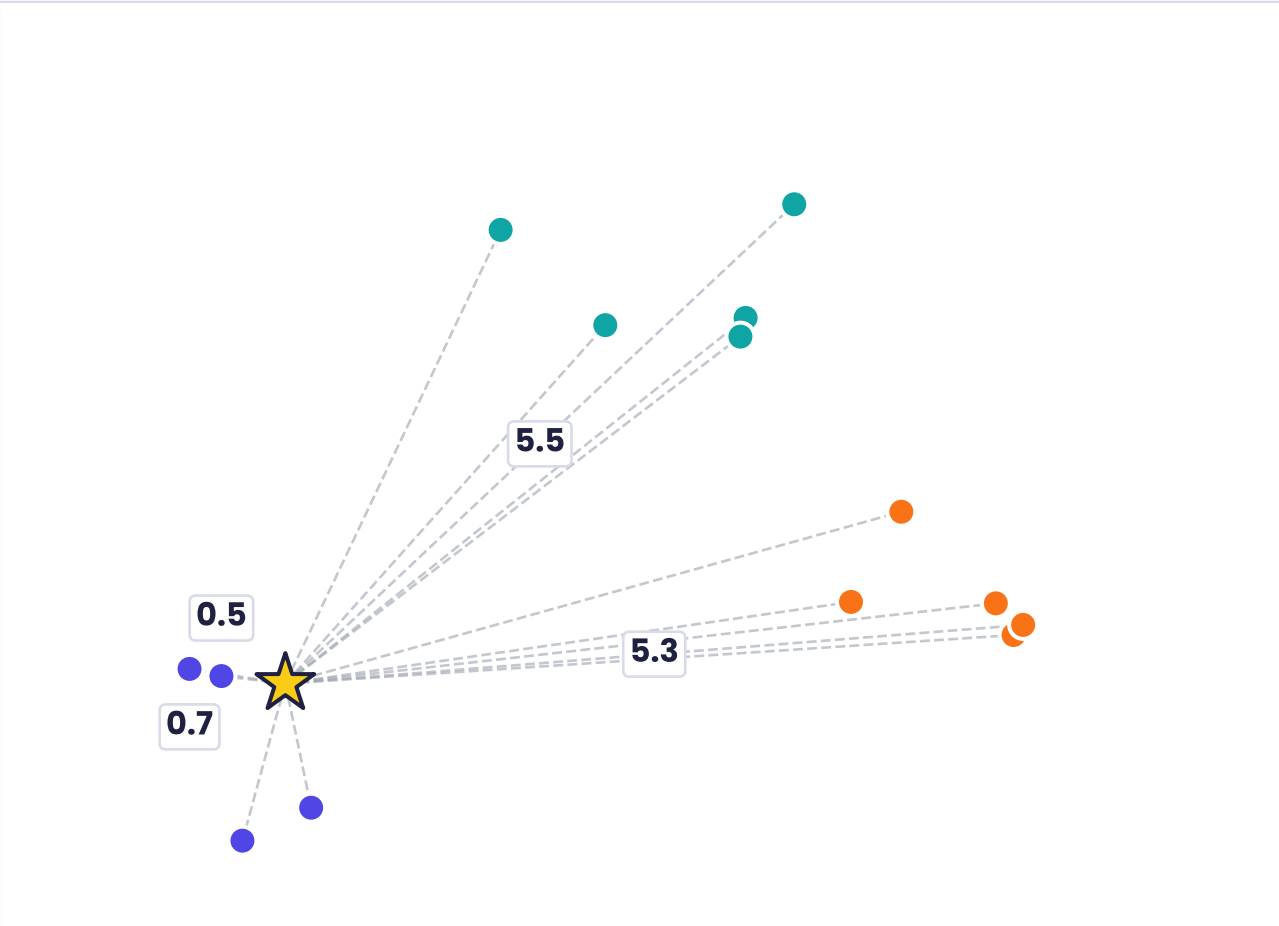
Pick one point: the star.

Measure its distance to every other point.

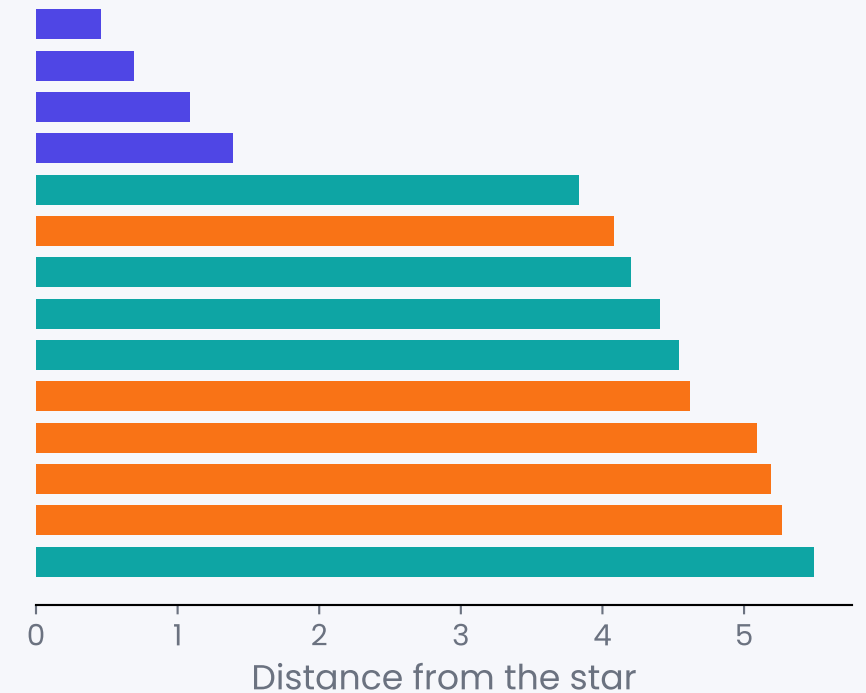
Small distance means very similar. Large distance means very different.

In real data, this distance is measured across all 64 numbers.

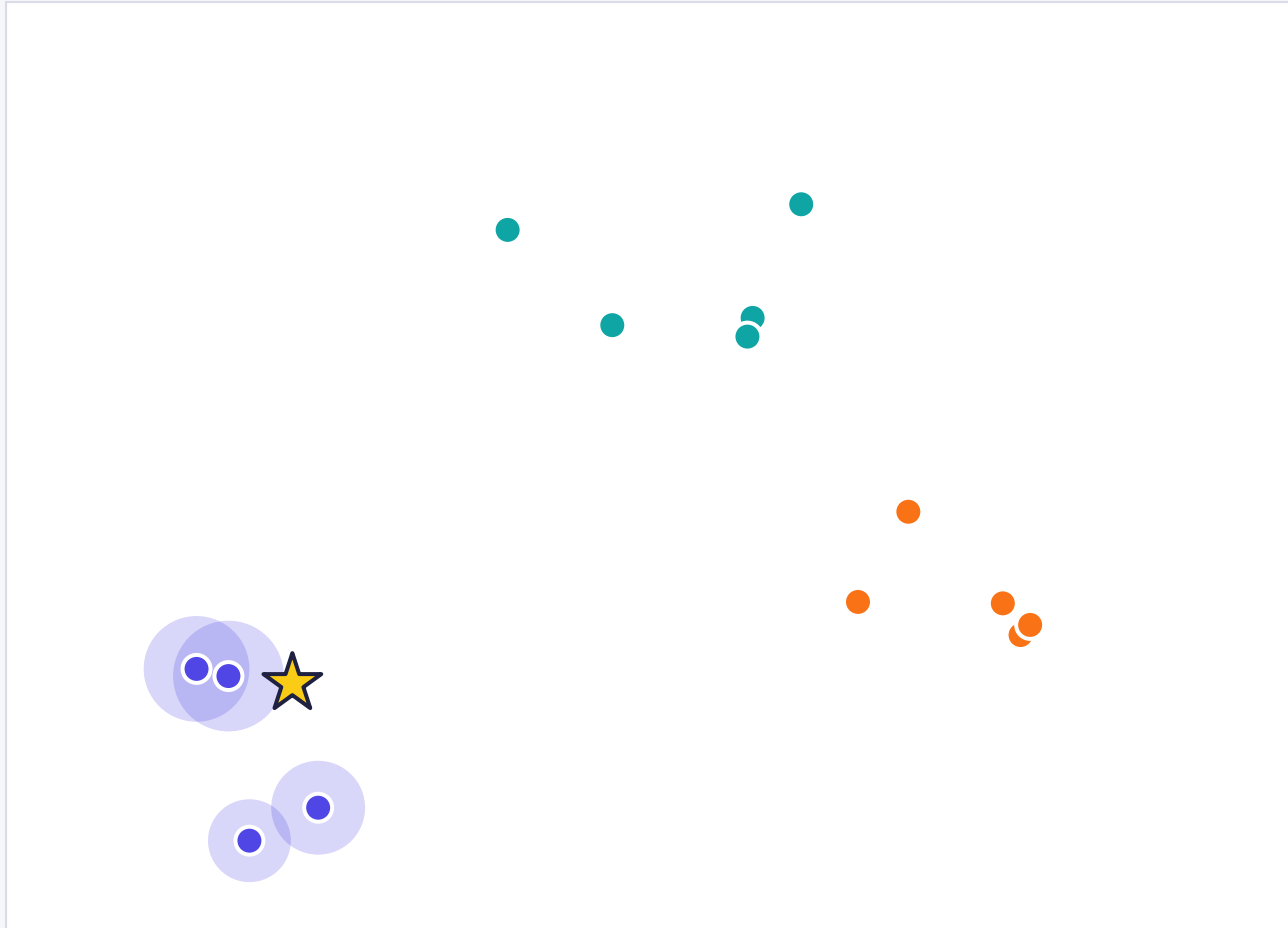
Step 1: Measure distances



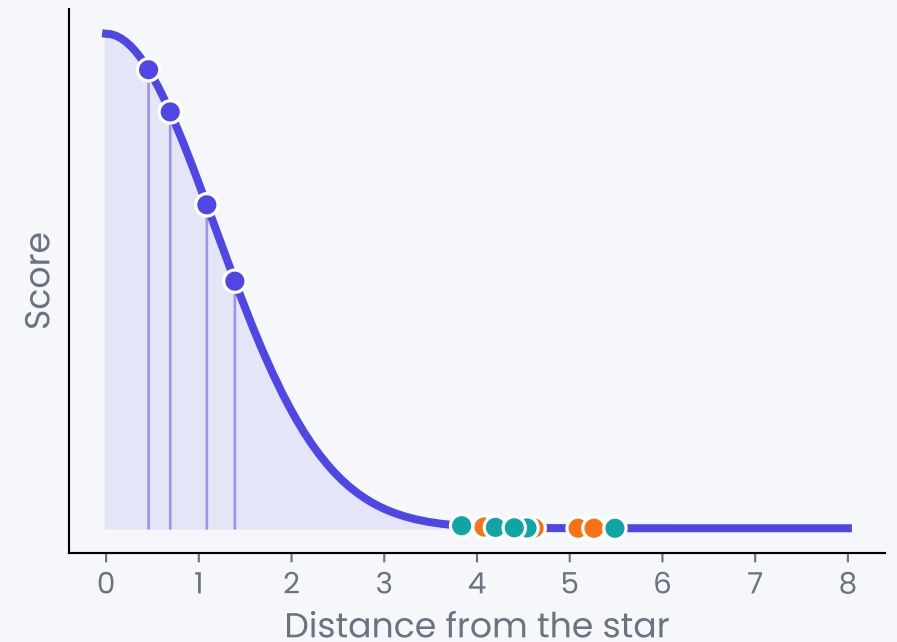
Do this for every point in the data, not only the star.



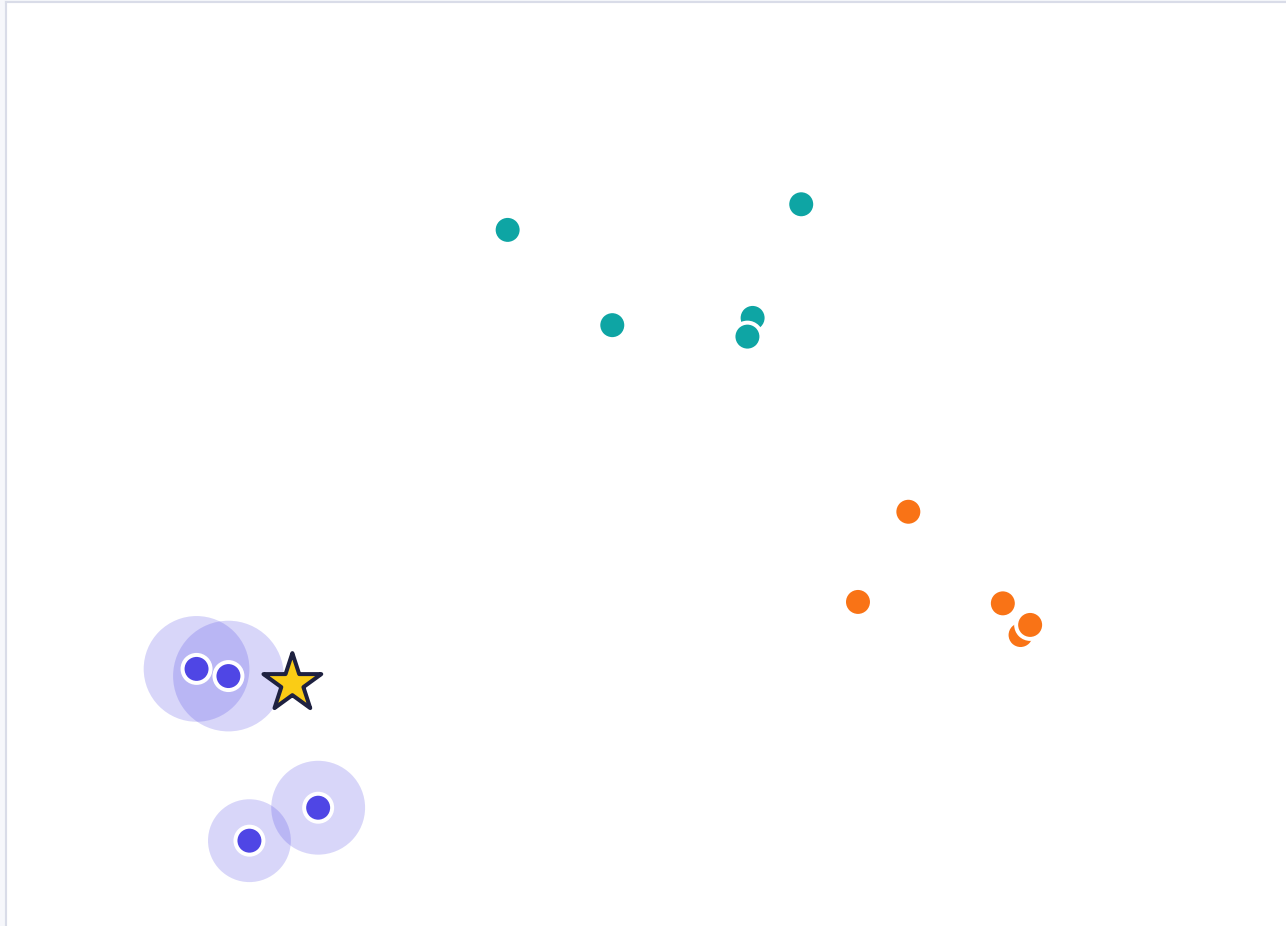
Step 2: Turn distances into probabilities



A bell curve turns each distance into a score. Close means high. Far means almost zero.

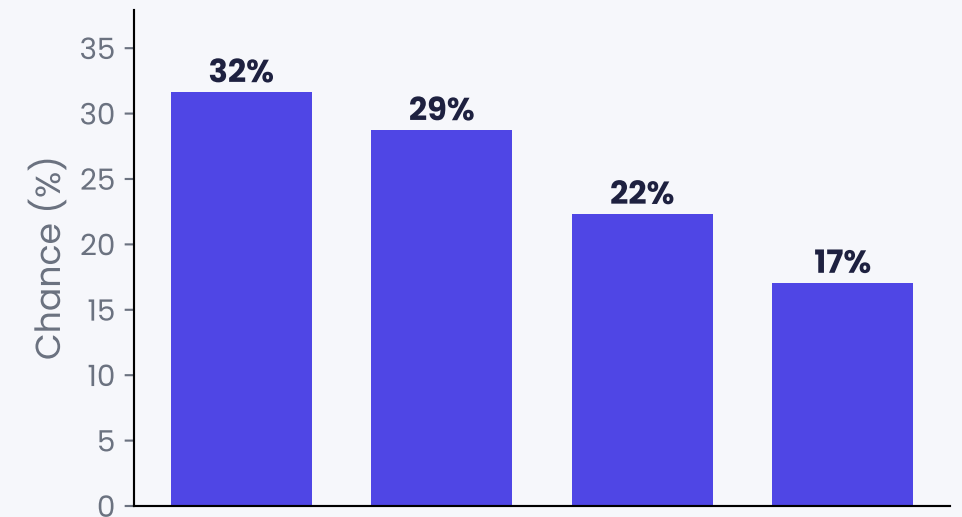


Step 2: Turn distances into probabilities



Scale the scores so they add up to 100%.

Each number is the chance that the star picks that point as a neighbor.

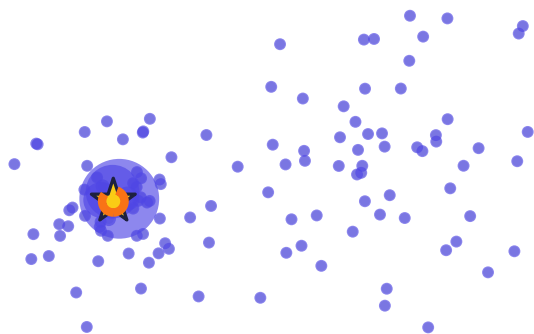


The star's 4 closest points. All others are near 0%.

Step 3: Choose how many neighbors matter

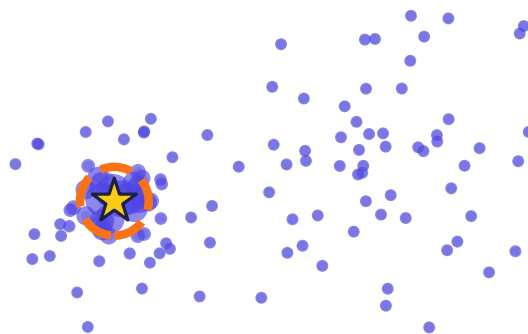
Perplexity is roughly how many close friends each point pays attention to.

Perplexity 3



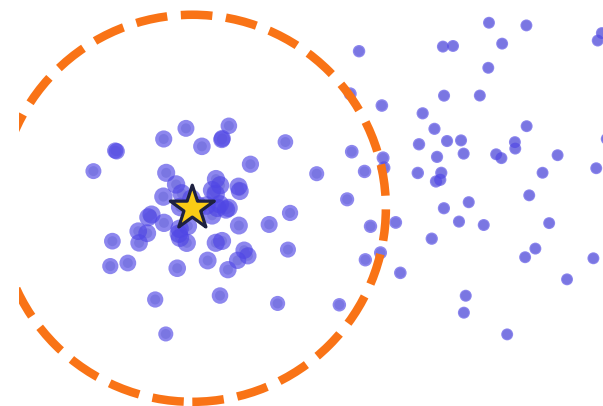
Very local view

Perplexity 20



Balanced view

Perplexity 70

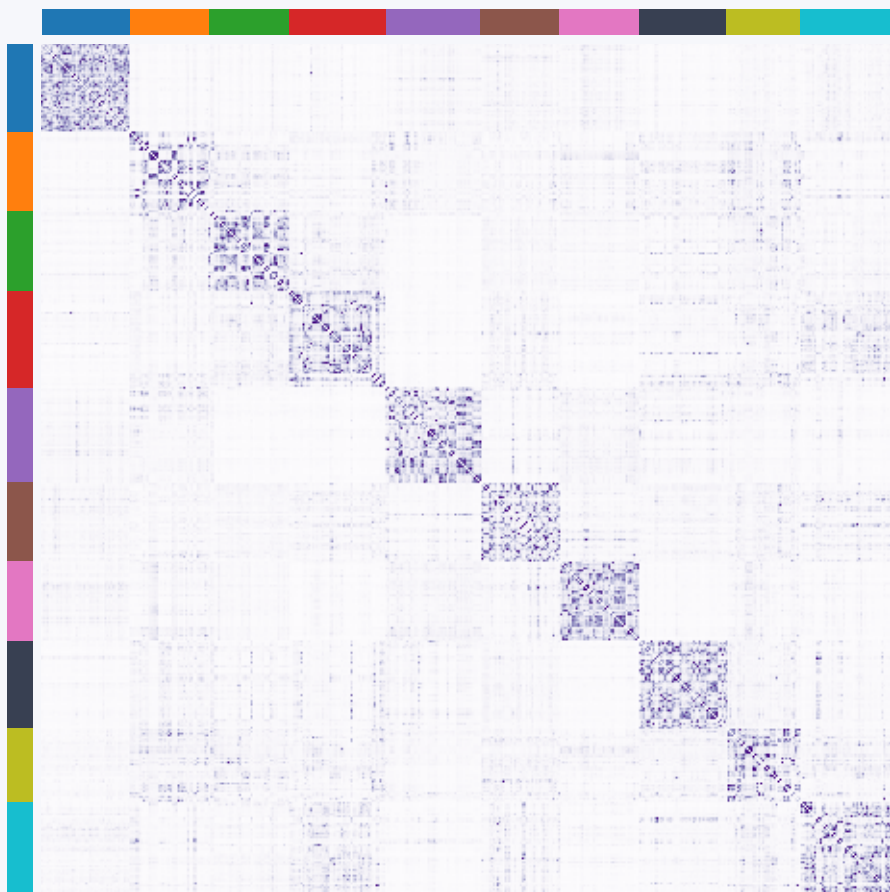


Wide view

Small: only the closest neighbors count. Large: many points count. A common choice is 5 to 50.

Step 4: Build the friendship table

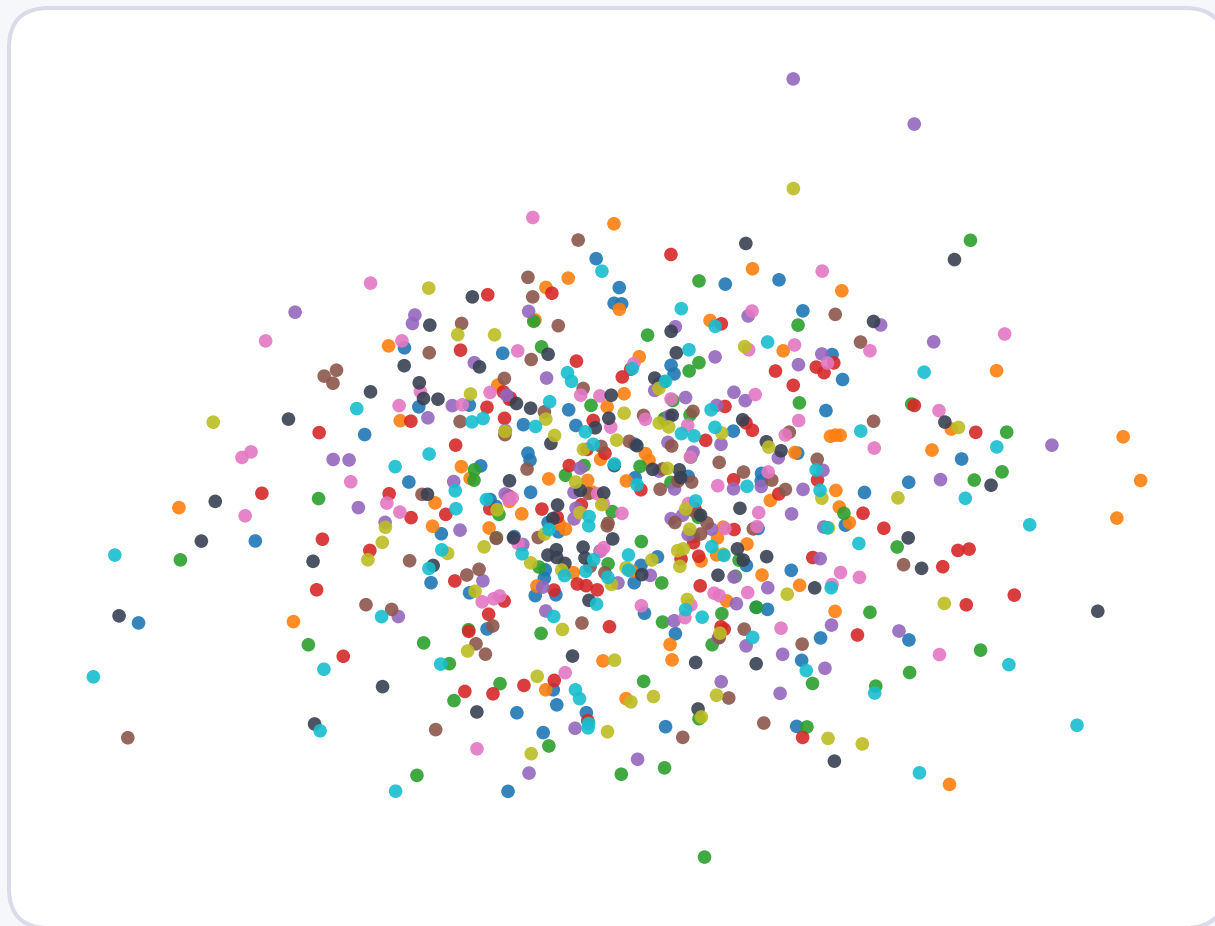
700 pictures (rows) x 700 pictures (columns)



- Every row and every column is one picture.
- Bright means strong friends. Dark means strangers.
- The same digit means friends, so bright squares appear along the diagonal.
- We average A-to-B and B-to-A so the table is fair both ways.



Step 5: Start with a random 2D map



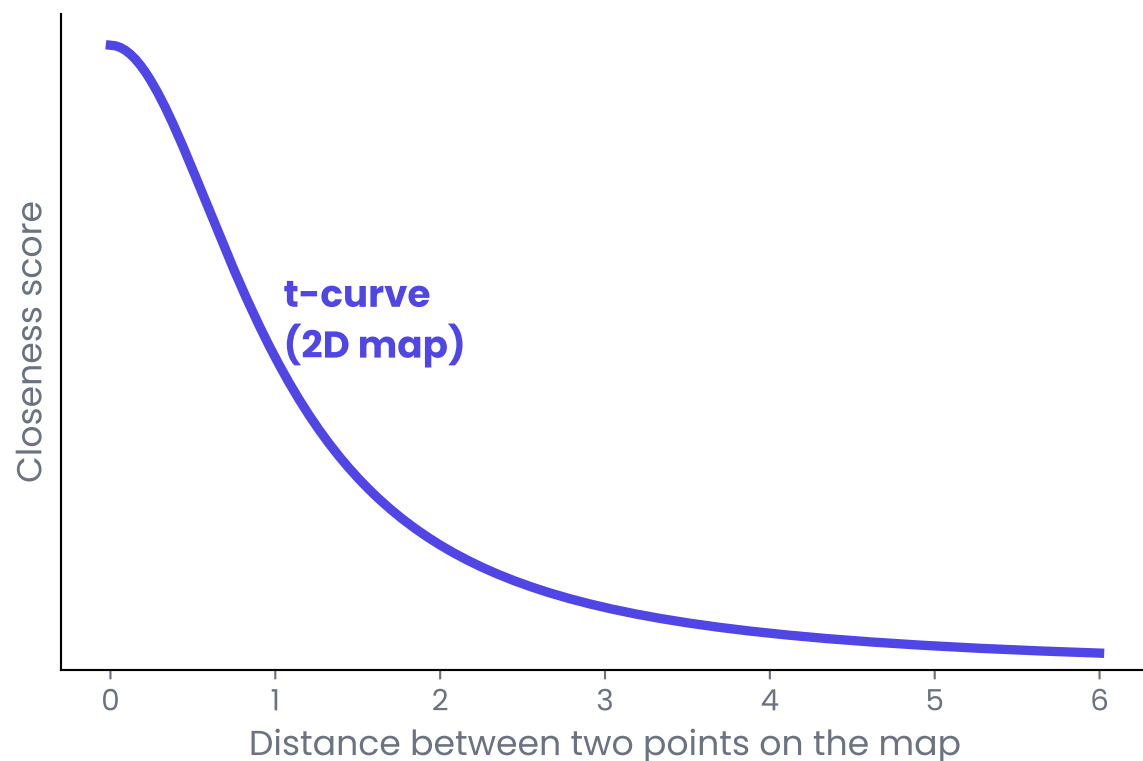
Now we place the same 700 pictures at random spots on a flat 2D sheet.

This is only a starting guess, so it is messy.

The colors are only for us. t-SNE does not know which digit is which.

The next steps will move these points until friends sit together.

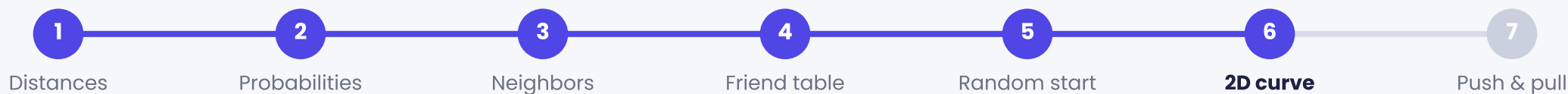
Step 6: Measure closeness on the map



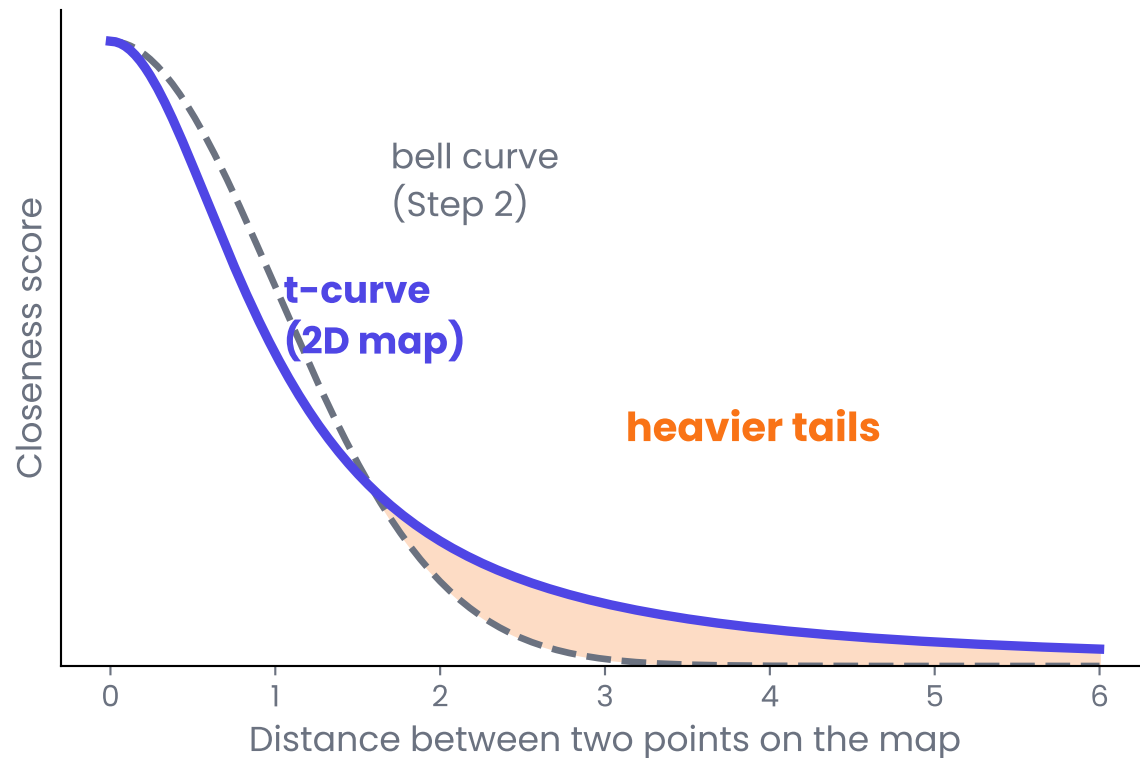
On the 2D map, we score closeness again.

This time we use a different curve, called the t-curve. This is where the t in t-SNE comes from.

Normalizing these scores gives a second table, called Q.



Step 6: Measure closeness on the map



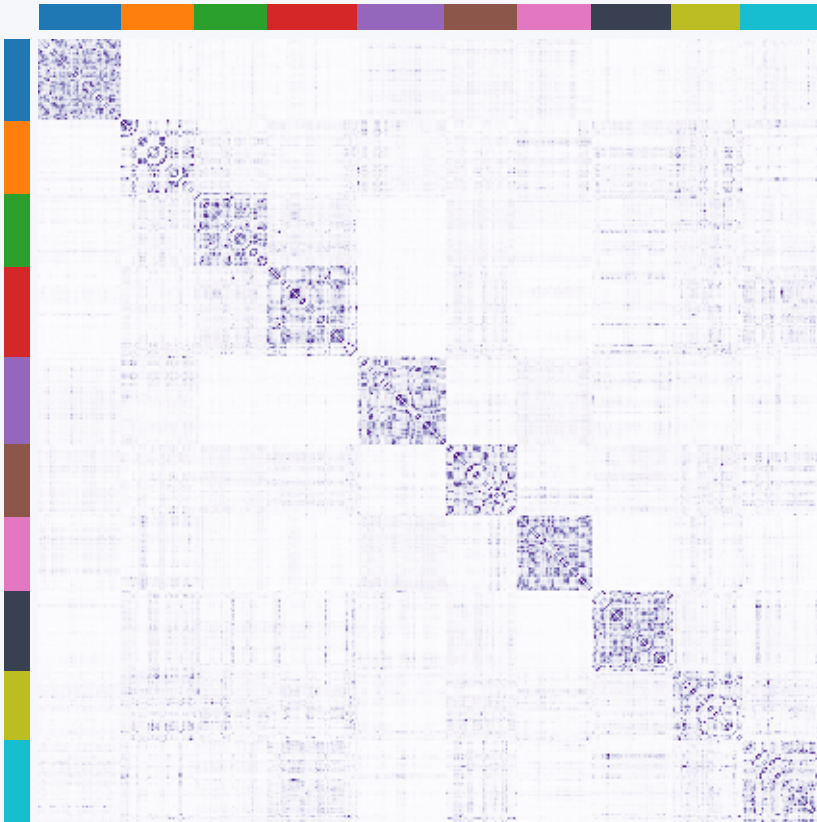
The t-curve has heavy tails.

Points that are far away still keep a small score, so they do not need to be squeezed in.

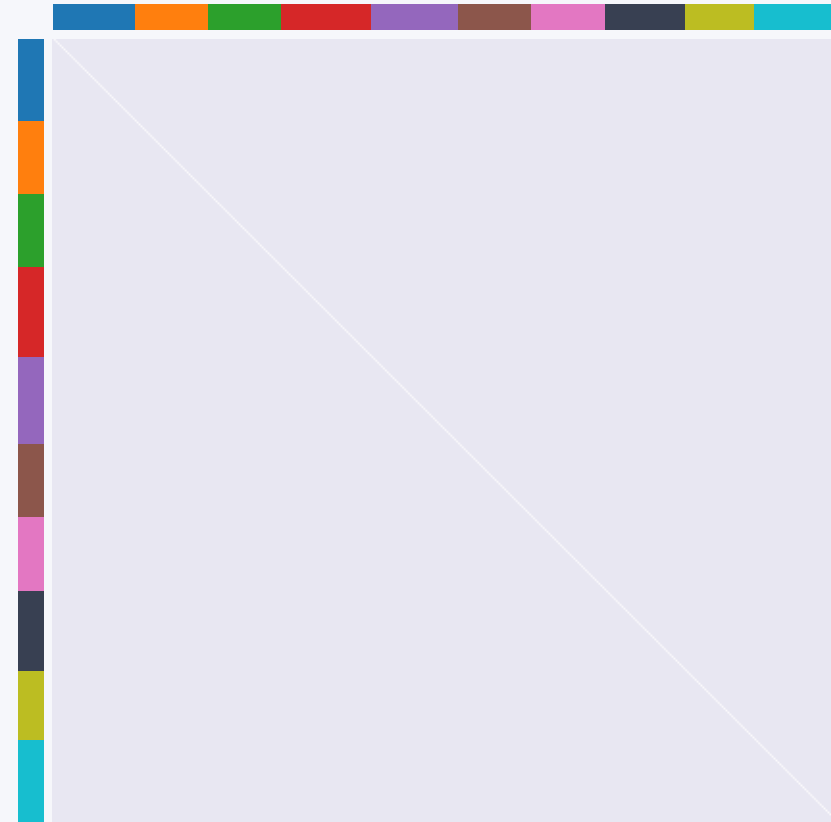
This prevents the crowding problem, where everything piles up in the middle. Groups get room to separate.

Compare the two tables

Friend table P (high dimensions)



Map table Q (2D map)



VS

Mismatch score: 3.06 (lower is better)



Step 7: Push and pull, then repeat

Pull



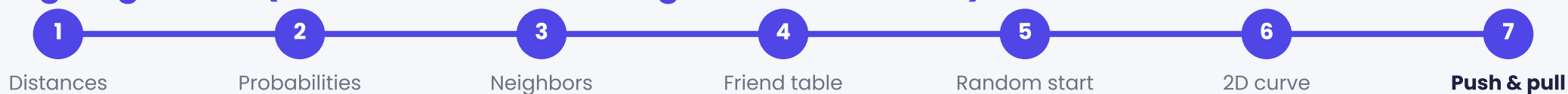
Friends in table P, but far apart on the map: pull them closer.

Push

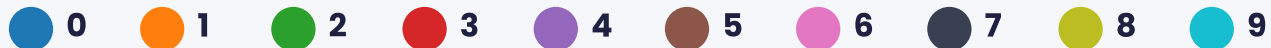
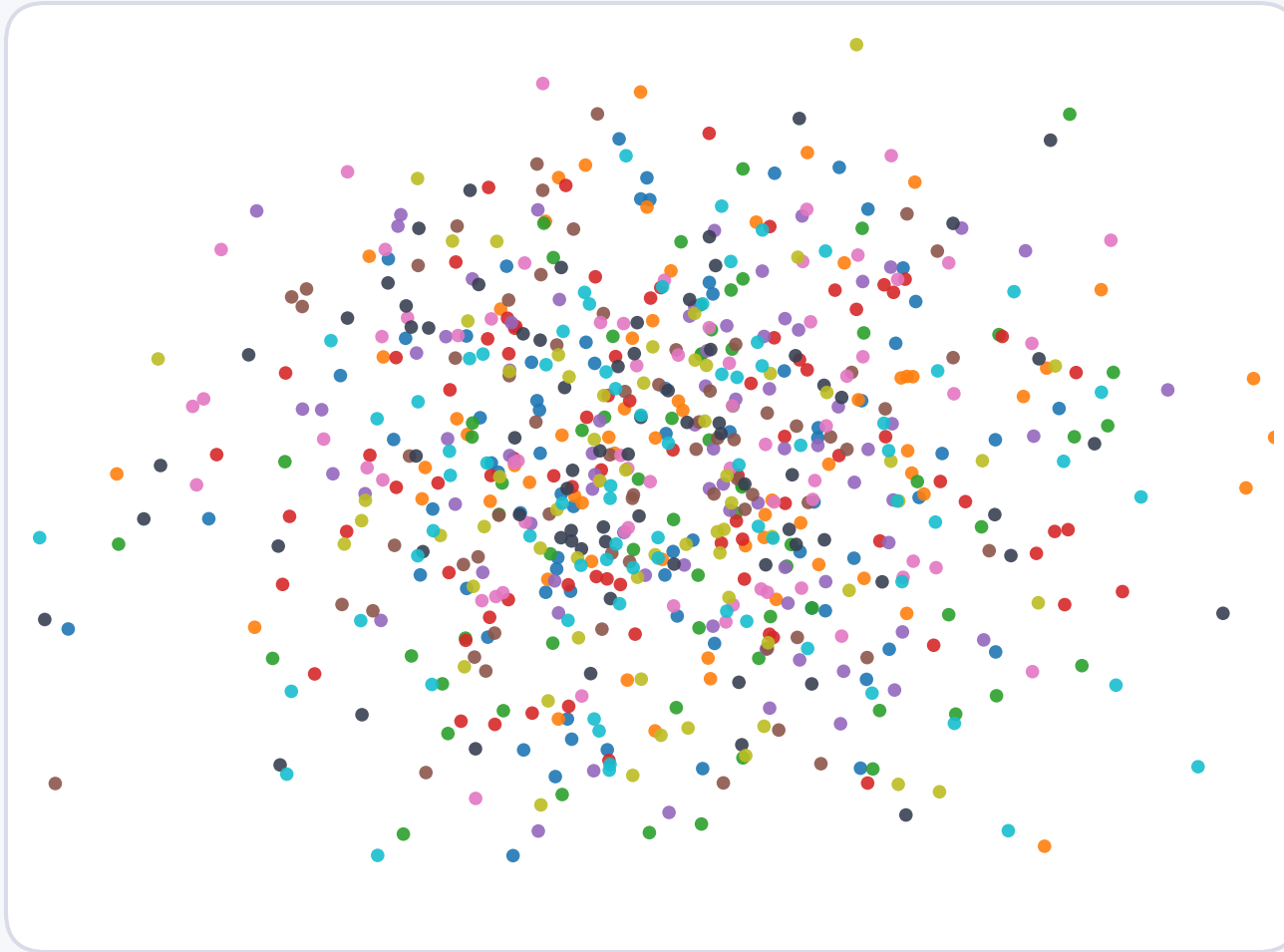


Strangers in table P, but too close on the map: push them apart.

Every round nudges every point a little. We repeat about 1,000 times, and the mismatch score keeps going down. (This method is called gradient descent.)



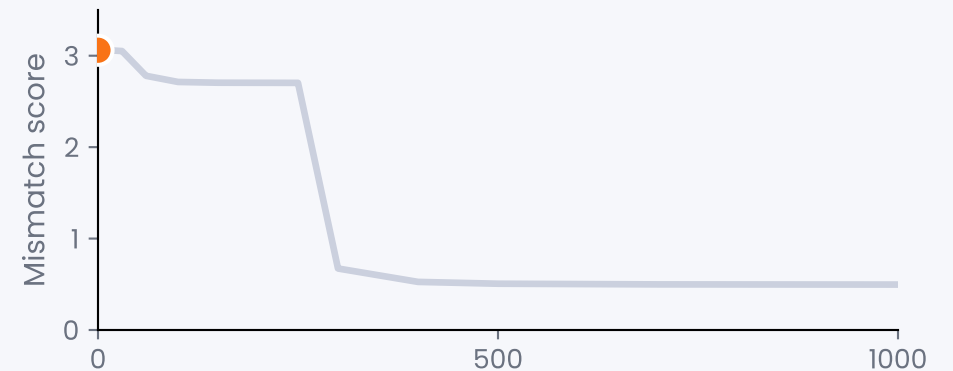
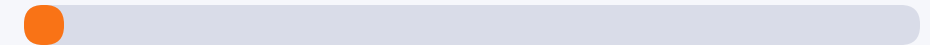
Simulation: watch the map form



Round

0

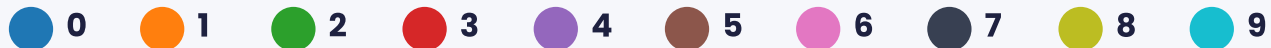
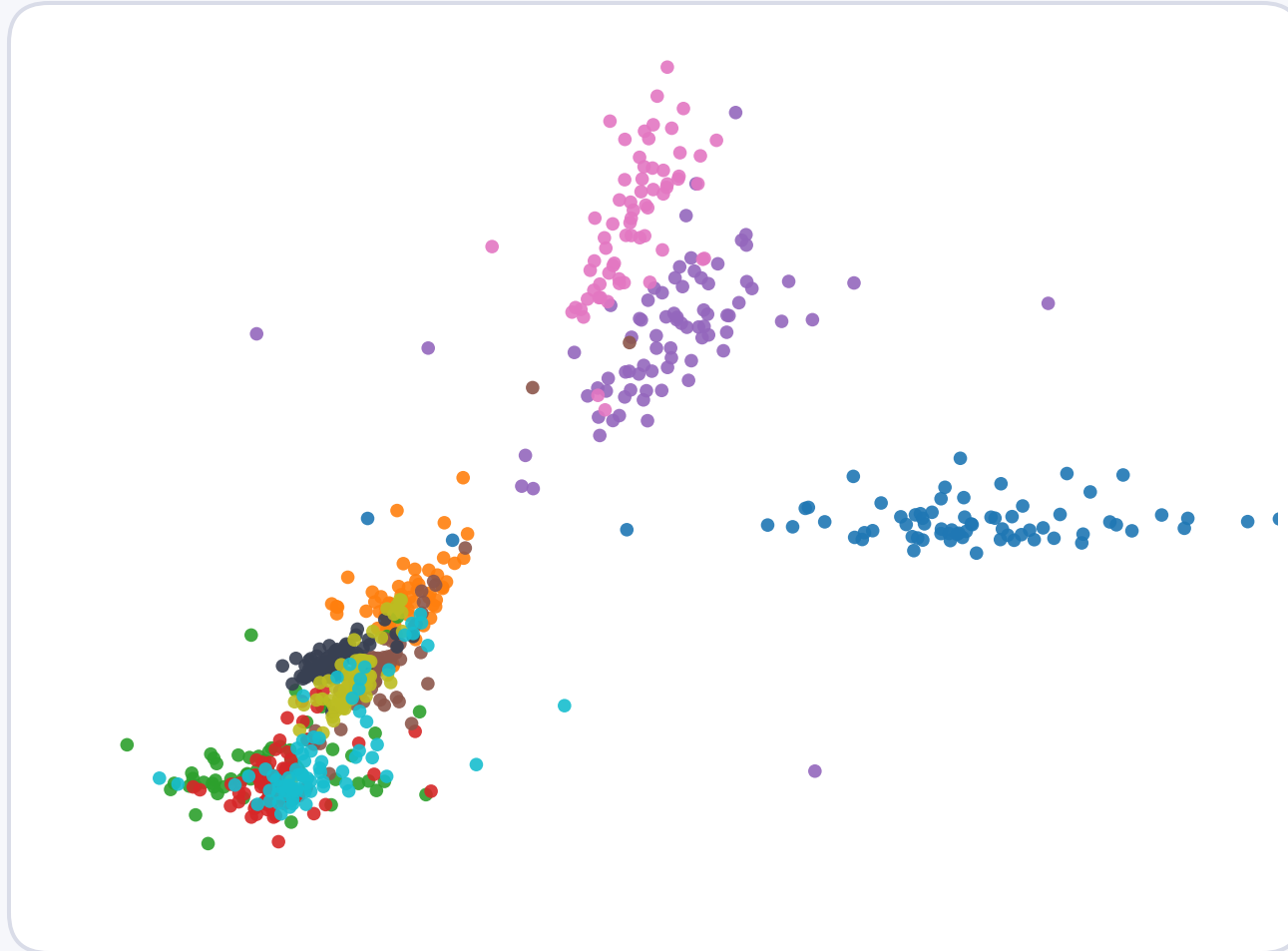
of 1,000



Random start. Groups are mixed.

Mismatch score is high.

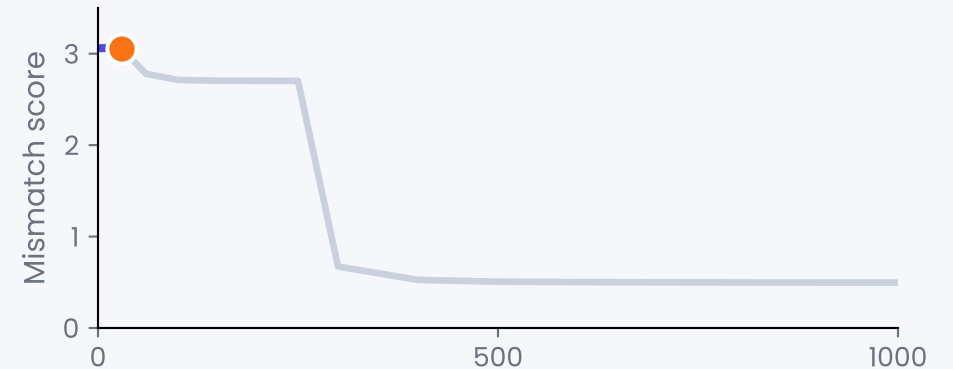
Simulation: watch the map form



Round

30

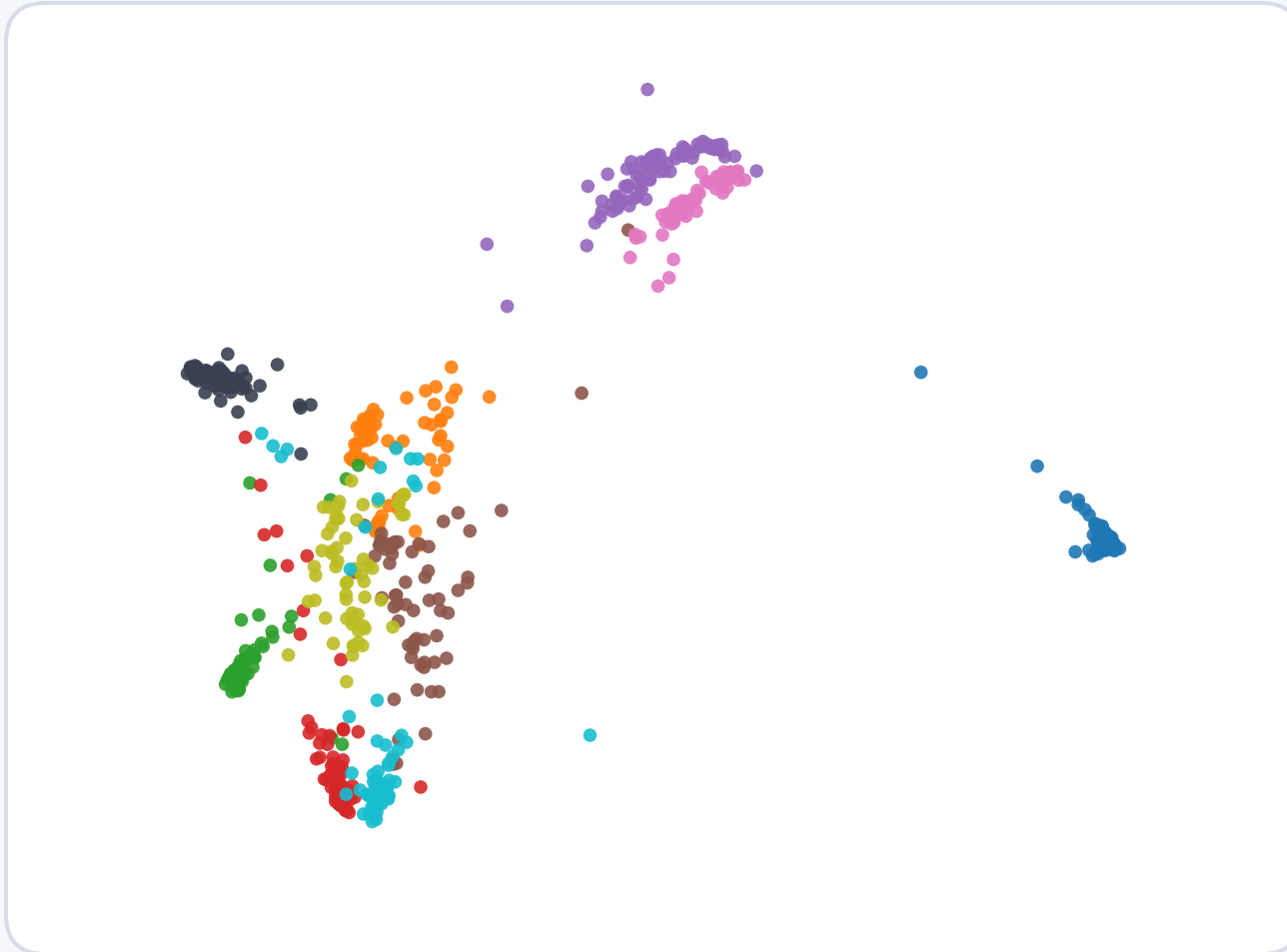
of 1,000



Points feel the pull.

Friends begin to drift together.

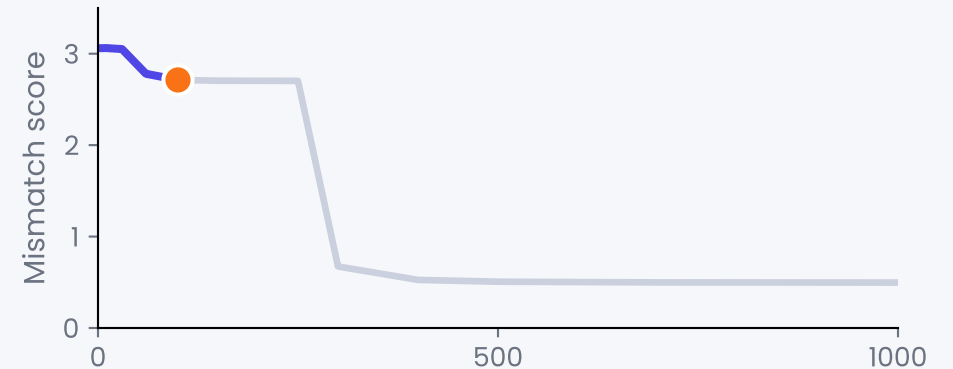
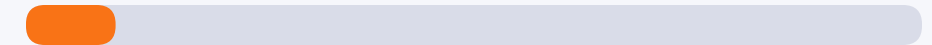
Simulation: watch the map form



Round

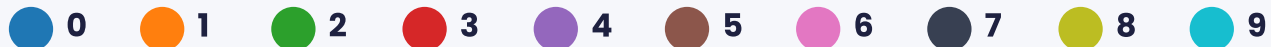
100

of 1,000



Groups start to form.

For the first 250 rounds, the pull is made extra strong (early exaggeration).



Simulation: watch the map form

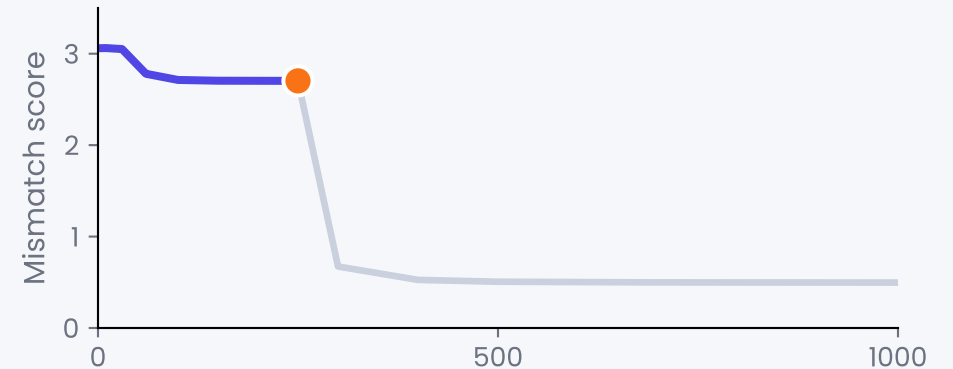


0 1 2 3 4 5 6 7 8 9

Round

250

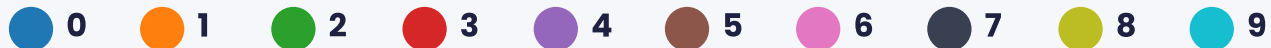
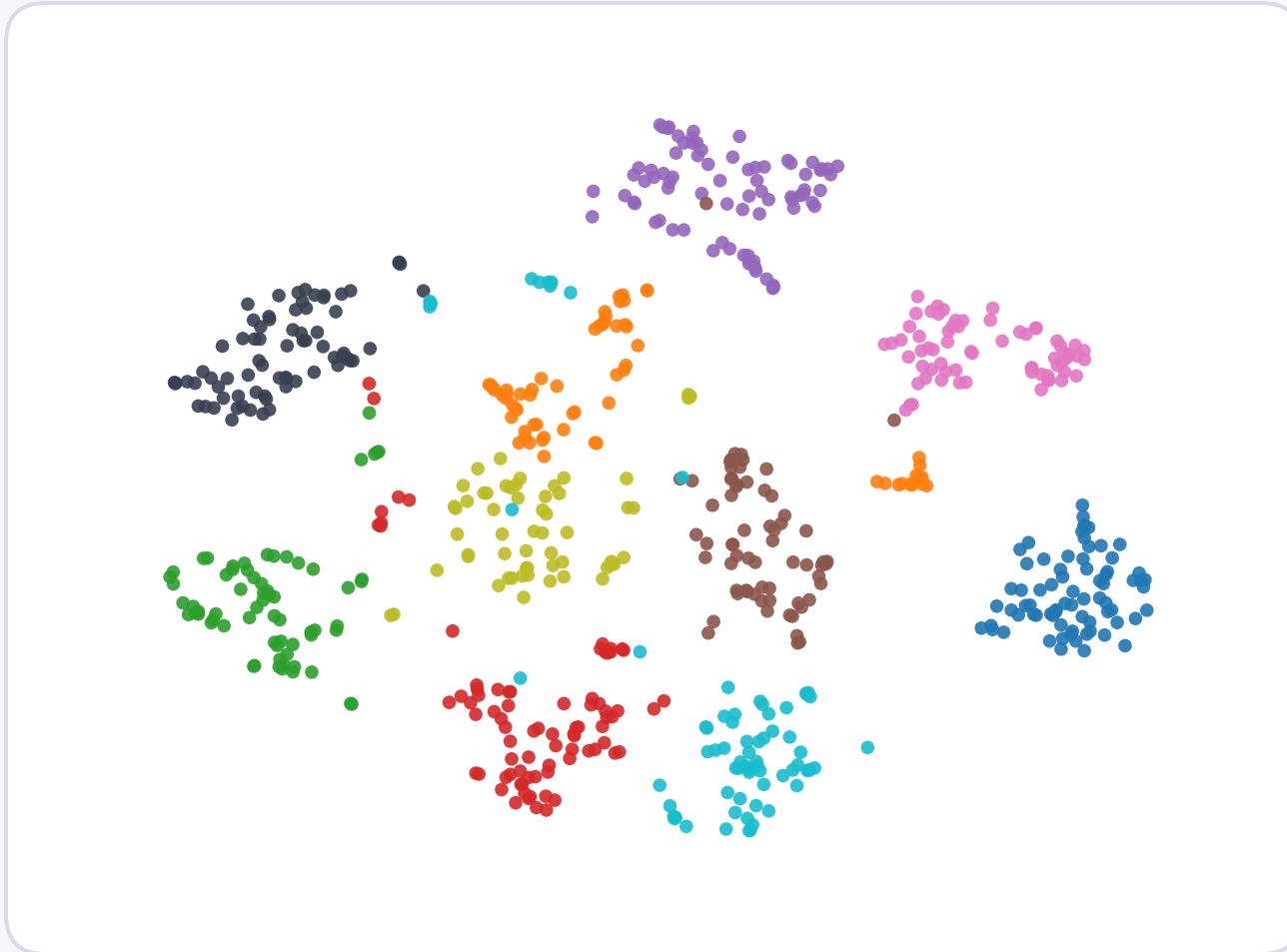
of 1,000



Groups are tight but squeezed.

Next, the extra pull is switched off.

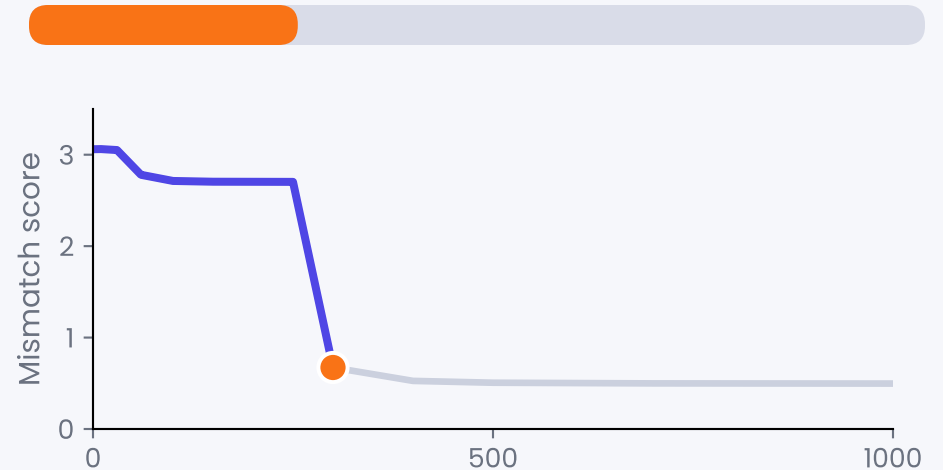
Simulation: watch the map form



Round

300

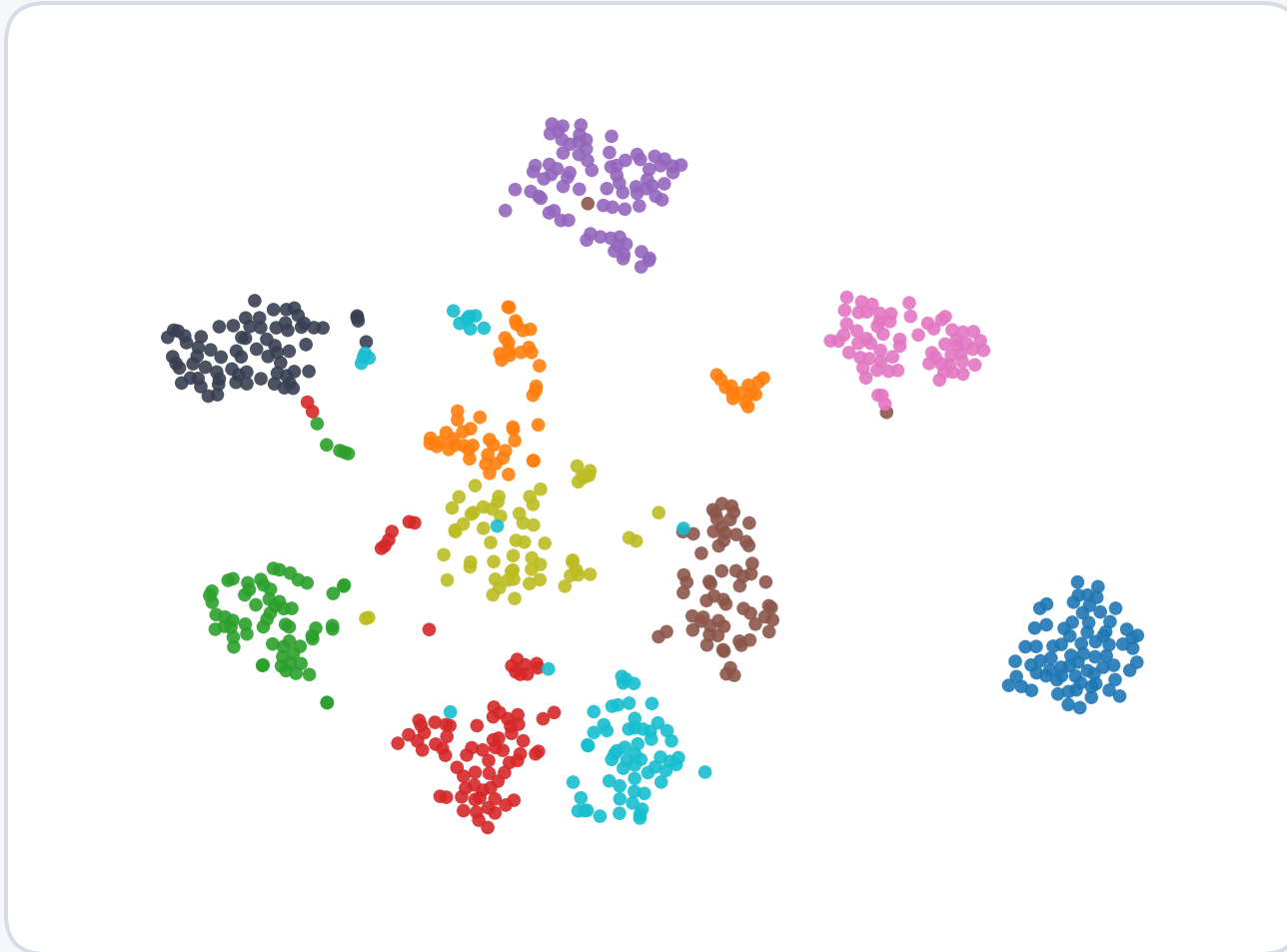
of 1,000



Groups spread out and separate.

The score drops quickly.

Simulation: watch the map form

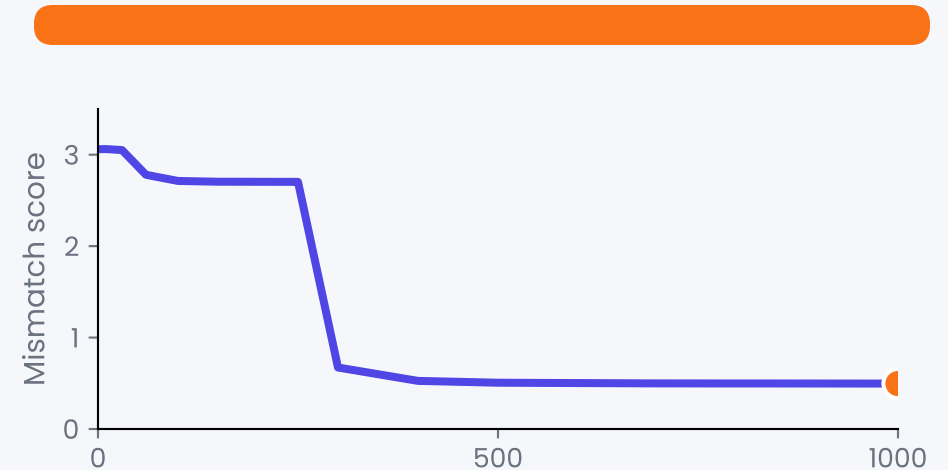


0 1 2 3 4 5 6 7 8 9

Round

1,000

of 1,000



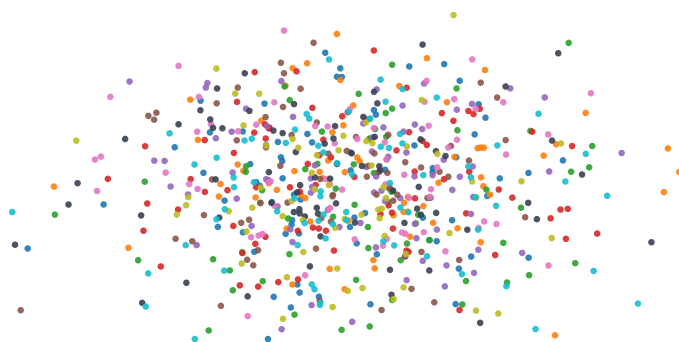
Settled: ten groups.

One group for each digit.

The whole run at a glance

Round 0

score 3.06



Round 30

score 3.05



Round 100

score 2.71



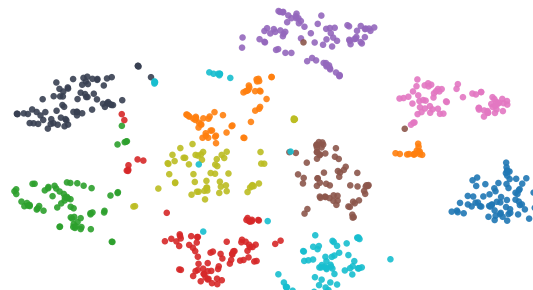
Round 250

score 2.70



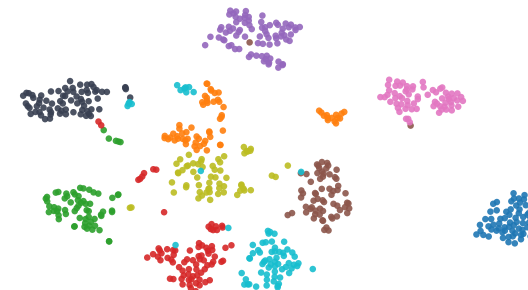
Round 300

score 0.67



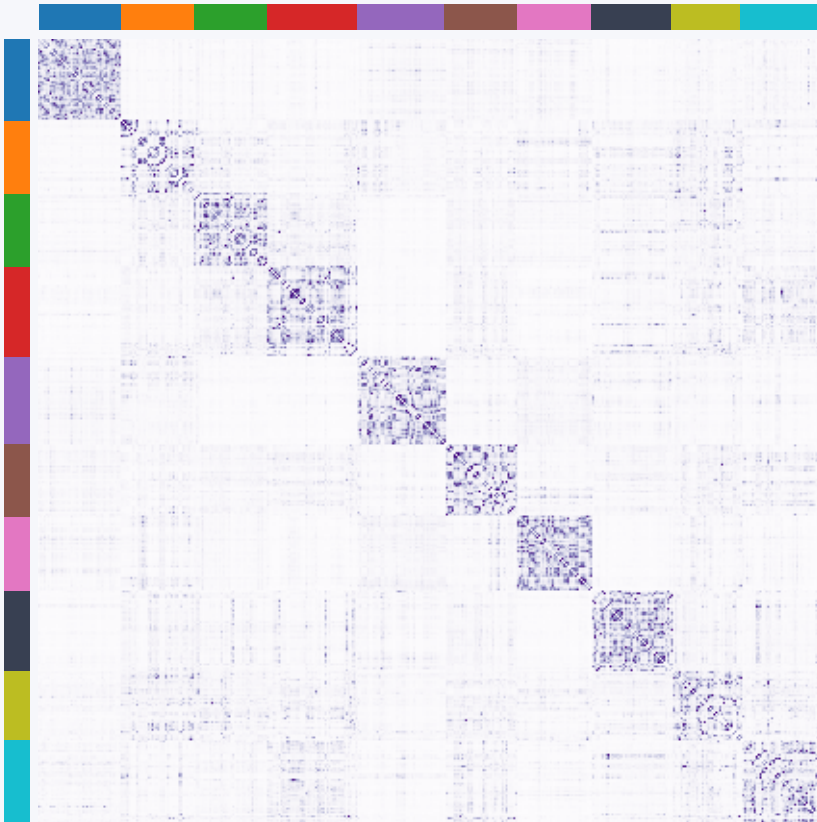
Round 1,000

score 0.50

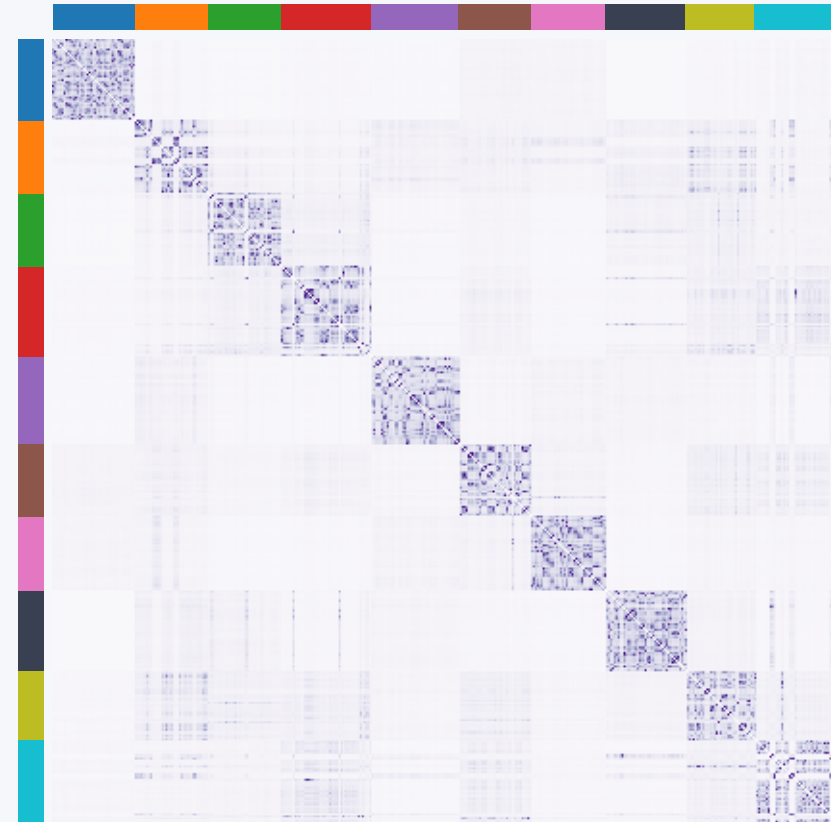


After 1,000 rounds: the tables match

Friend table P (high dimensions)



Map table Q (2D map)



vs

Mismatch score: 0.50 (much lower)



How to read a t-SNE map



You can trust

Points close together are similar.

Well-separated groups are usually really different.

Odd points far from the rest may be outliers.



Be careful

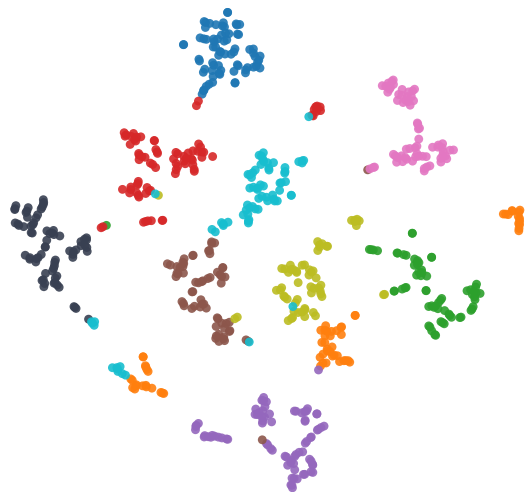
The size of a group can be misleading.

The gap between groups may not mean much.

The x and y axes have no meaning, and each run can look different.

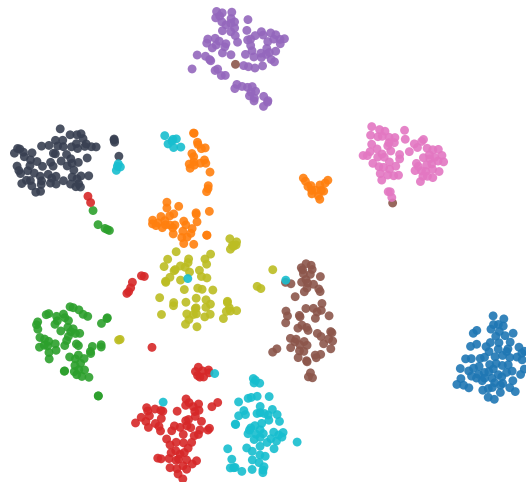
Perplexity changes the picture

Perplexity 5



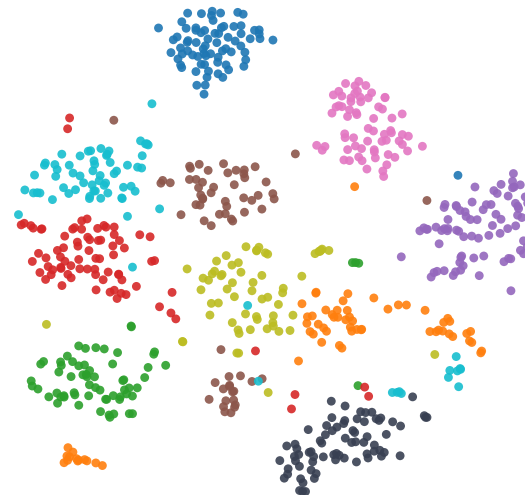
Too small: groups break into pieces.

Perplexity 30



Balanced: clear, clean groups.

Perplexity 100



Too large: groups start to blend.

Try several values, then compare. Do not trust a single picture.

Practical tips

1

Try several perplexity values.

Groups that appear every time are the ones you can trust.

2

Run it more than once.

The random start changes the picture a little each time.

3

Shrink big data first with PCA.

PCA is still useful as a fast first step, then t-SNE draws the map.

4

Use it to see, not to measure.

Use t-SNE for exploring. Do not read exact distances from it.

Recap: t-SNE in 7 steps

- 1 Measure distances between points.
- 2 Turn distances into probabilities.
- 3 Pick the perplexity (how many neighbors matter).
- 4 Build the friendship table P.
- 5 Place points randomly on a 2D map.
- 6 Score closeness on the map with the t-curve (table Q).
- 7 Push and pull until Q matches P.

In one sentence: t-SNE draws a 2D map in which points that are neighbors in many dimensions stay neighbors on the page.

Check your understanding

1

Why can PCA fail on a rolled-up sheet of data?

2

What does perplexity control?

3

Why do we use a t-curve on the 2D map?

Answers

1

Why can PCA fail on a rolled-up sheet of data?

It only uses straight-line shadows, so different layers land on top of each other.

2

What does perplexity control?

Roughly how many close neighbors each point pays attention to.

3

Why do we use a t-curve on the 2D map?

Its heavy tails give groups room to separate, which avoids the crowding problem.

The background is a dark blue field filled with numerous small, semi-transparent dots in various colors including purple, orange, green, red, and light blue. These dots are arranged in several distinct, irregular clusters that vaguely resemble the shapes of continents or abstract organic forms. The text 'Thank you!' is centered in a large, white, sans-serif font, with the word 'Thank' on the top line and 'you!' on the bottom line. Below this, the word 'Questions?' is centered in a smaller, light blue, sans-serif font.

Thank you!

Questions?